

MICROBIOLOGY – SHORT EXAM NOTES

Epidemiology of Infectious Disease • Normal Flora of Human Body • Sample Collection, Transport & Storage
(Ananthanarayan & Paniker)

1. EPIDEMIOLOGY OF INFECTIOUS DISEASE

Infection and immunity involve an interaction between the **host** and the **microorganism**. **Saprophytes** live on dead or decaying matter (soil, water) and rarely cause disease, except **opportunistic infection** in devitalised hosts (e.g. *B. subtilis*). **Parasites** multiply in hosts and are either **pathogens** (produce disease) or **commensals** (live in harmony without harm). Many commensals are **facultative pathogens** that cause disease when host resistance falls. **Infection** is the lodgement and multiplication of a parasite in the host; it does not always lead to **disease**.

Classification of Infections

Type	Meaning
Primary infection	Initial infection with a parasite
Re-infection	Subsequent infection by the same parasite
Secondary infection	New parasite infects a host whose resistance is lowered by an existing disease
Focal infection (focal sepsis)	Localised infection (tonsils, appendix) producing generalised effects
Cross-infection	New infection from another host/external source in a patient already ill
Nosocomial infection	Cross-infection acquired in hospital
Iatrogenic infection	Physician-induced, due to procedures
Endogenous / Exogenous	Source within / outside the host's body
Inapparent (subclinical)	Clinical effects not apparent
Atypical	Characteristic features of the disease absent
Latent	Parasite hidden in tissues; reactivates when resistance falls

Sources of Infection

- **Humans** – the **commonest source**; from a **patient** or a **carrier**. A carrier harbours the pathogen without ill effect: **healthy carrier** (never had the disease), **convalescent carrier** (continues after recovery), **temporary** (<6 months) or **chronic** (years/life-long), **contact carrier** (acquires from a patient), **paradoxical carrier** (acquires from another carrier).
- **Animals** – act as **reservoir hosts**; diseases transmitted from animals to humans are **zoonoses** – plague (rats), rabies (dogs), toxoplasmosis (cats), hydatid disease (dogs).
- **Insects (vectors)** – cause **arthropod-borne diseases**. **Mechanical vectors** only carry the pathogen (housefly – typhoid, dysentery); **biological vectors** allow multiplication/development (*Aedes* – yellow fever, *Anopheles* – malaria). The time from entry into the vector to it becoming infective is the **extrinsic incubation period**. Some insects are also reservoir hosts (ticks), maintaining infection by **transovarial** passage.
- **Soil** – tetanus spores (viable for decades), *Histoplasma*, *Nocardia*, roundworm, hookworm.
- **Water** – contamination (cholera vibrio, hepatitis virus) or aquatic vectors (cyclops in guinea worm).
- **Food** – external contamination (staphylococcal food poisoning) or pre-existing infection in meat (salmonellosis).

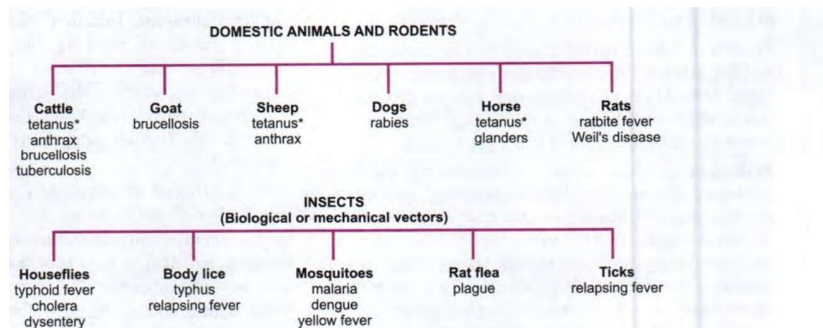


Fig. 8.1 Possible sources of infection (from the book)

Methods of Transmission

- **Contact** – **direct** (syphilis, gonorrhoea) or **indirect** through **fomites** – inanimate objects like clothing, pencils, toys, towels (diphtheria, trachoma).
- **Inhalation** – respiratory infections (influenza, tuberculosis) spread by droplets; small droplets dry into **droplet nuclei (1–10 µm)** that remain suspended in air.
- **Ingestion** – food/water/hands: **waterborne** (cholera), **foodborne** (food poisoning), **handborne** (dysentery).
- **Inoculation** – directly into tissues: tetanus spores in wounds, rabies by dog bite, arboviruses by insects; **hepatitis B and HIV** via blood transfusion and contaminated needles.
- **Insects** – mechanical or biological vectors.
- **Congenital (vertical)** – across the placenta: congenital syphilis; rubella in the first trimester causes malformations (**teratogenic infection**).
- **Iatrogenic and laboratory infections** – injections, lumbar puncture, catheterisation, dialysis, transplants; laboratory workers handling infectious material.

Pathogenicity and Virulence

Pathogenicity is the ability of a **species** to cause disease; **virulence** is the same property of a **strain**. Increase in virulence is **exaltation** (serial passage in susceptible hosts); decrease is **attenuation**. Virulence depends on: **adhesion** (adhesins, fimbriae), **invasiveness**, **toxigenicity** (exotoxins and endotoxins), **communicability**, capsules and surface antigens (resist phagocytosis), enzymes (**coagulase**, **fibrinolysin**, **hyaluronidase**, **leucocidins**), **plasmids** and **bacteriophages** (diphtheria toxin gene), **biofilms**, **infecting dose** (MID, MLD, ID₅₀, LD₅₀) and **route of infection**.

Exotoxins vs Endotoxins

Exotoxins	Endotoxins
Proteins	Lipopolysaccharides (lipid A)
Heat labile	Heat stable
Actively secreted; diffuse out	Part of Gram-negative cell wall; released on lysis
Specific action and tissue affinity	Non-specific, similar for all
Active in minute doses	Active only in large doses
Highly antigenic; converted to toxoid	Weakly antigenic; cannot be toxoided
Neutralised by antibody	Neutralisation ineffective

Types of Infectious Diseases

Infections may be **localised** (superficial or deep) or **generalised**. **Bacteremia** is circulation of bacteria in blood (transient bacteremia occurs even on brushing teeth). **Septicemia** – bacteria circulate and **multiply** in blood, form toxins and cause high swinging fever. **Pyemia** – septicemia by pyogenic bacteria with **multiple abscesses** in internal organs.

According to spread in the community:

- **Endemic** – constantly present in an area (typhoid in India).
- **Epidemic** – spreads rapidly, involving many people in an area at the same time (influenza in winter).
- **Pandemic** – an epidemic spreading over many parts of the world (influenza, cholera, plague).
- **Prosodemic** – slow, creeping epidemic spread person-to-person (cerebrospinal fever), unlike **explosive** waterborne outbreaks (cholera, hepatitis).

★ EXAM FOCUS

- Short answers: **methods of transmission, carriers, exotoxin vs endotoxin, bacterial virulence factors.**
- Define: zoonosis, fomite, nosocomial, iatrogenic, extrinsic incubation period, endemic/epidemic/pandemic.

2. NORMAL MICROBIAL FLORA OF THE HUMAN BODY

The normal flora are more or less constant for each species and are of two kinds. **Resident flora** form a constant population that cannot be permanently removed and prevent colonisation by other organisms. **Transient flora** are present only temporarily and may be pathogenic or non-pathogenic.

Role of Normal Flora

- **Beneficial:** prevent colonisation/invasion by pathogens; raise immune status against related antigens; gut flora synthesise **vitamin K and B vitamins**; produce antibacterial substances (**colicins**); small amounts of endotoxin stimulate the alternative complement pathway.
- **Harmful:** become pathogenic when host defences fail (**opportunistic**); cause **confusion in diagnosis**; penicillinase producers interfere with therapy; oral streptococci cause **dental caries**; abnormal multiplication can cause enteritis and endotoxic shock.

In hospitals, the flora shift towards **antibiotic-resistant staphylococci**; recolonisation with harmless penicillin-sensitive *S. aureus* strain **502A** has been used. New army recruits show increased carriage of meningococci and group A streptococci.

Flora at Different Sites

Site	Normal flora
Skin	Diphtheroids (propionibacteria), staphylococci, aerobic spore bearers, <i>S. viridans</i> , <i>E. faecalis</i> , coliforms, <i>Proteus</i> , non-pathogenic mycobacteria, <i>Candida</i> , <i>Pityrosporum ovale</i> . Hair harbours <i>S. aureus</i> .
Conjunctiva	Relatively free (flushing by tears); diphtheroids (<i>C. xerosis</i>), <i>Moraxella</i> , staphylococci, non-haemolytic streptococci.
Nose & nasopharynx	Corynebacteria, staphylococci, streptococci, <i>Haemophilus</i> , <i>Moraxella</i> . Infant nasopharynx sterile at birth; colonised in 2–3 days.

Site	Normal flora
Mouth & upper respiratory tract	Micrococci, lactobacilli, coliforms; anaerobes in gum pockets and tonsils – streptococci, fusiform bacilli, actinomyces, bacteroides, neisseria, <i>Candida</i> . Alpha-haemolytic streptococci dominate for life. Smaller bronchi and alveoli are sterile.
Stomach	Virtually sterile due to low pH (except after meals).
Intestine	Bacteria increase from duodenum (10^3 – 10^6 /g) to colon (10^{11}/g , 10–20% of fecal mass). Colon flora are 96–99% anaerobes (bacteroides, anaerobic streptococci, clostridia, lactobacilli) and 1–4% aerobes (coliforms, enterococci, <i>Proteus</i> , <i>Pseudomonas</i>). Breastfed infants: <i>L. bifidus</i> (99%).
Genitourinary tract	<i>M. smegmatis</i> in smegma (may confuse urine AFB tests); female urethra sterile or few cocci. Vagina: Döderlein's bacillus (lactobacillus) keeps pH acid in reproductive years (from glycogen under oestrogen); alkaline flora before puberty and after menopause.
Blood & tissues	Normally sterile ; commensals entering are quickly eliminated.

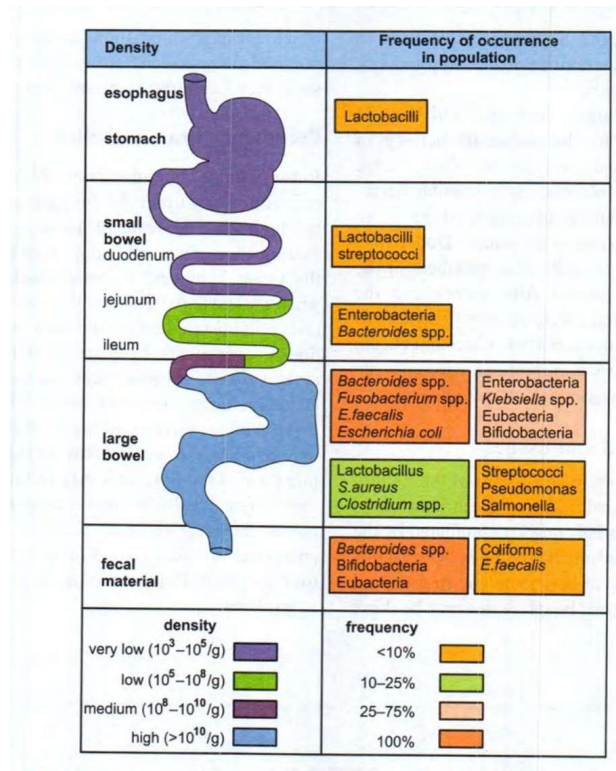


Fig. 65.1 Distribution and density of normal flora of the gastrointestinal tract (from the book)

Pseudomembranous Colitis

Indiscriminate use of broad-spectrum antibiotics (especially **clindamycin**) suppresses normal gut flora, allowing overgrowth of *Clostridium difficile* – a minor member of gut flora – causing life-threatening **pseudomembranous colitis**. Hence narrow-spectrum antibiotics in correct dose and duration are preferred.

★ EXAM FOCUS

- Short notes: **Normal flora of intestine, Pseudomembranous colitis, Role of normal flora.**
- Sterile sites: blood, CSF, alveoli, stomach (mostly), female urethra (mostly).

3. SAMPLE COLLECTION, TRANSPORTATION AND STORAGE

Diagnosis of infection depends on isolation and identification of the agent, detection of antigen, or demonstration of antibodies (**IgM – recent, IgG – past** infection). **"The laboratory report is only as good as the specimen that is sent."** The laboratory process has three steps:

1. **Pre-analytical (pre-test)** – specimen collection and transport.
2. **Analytical** – processing in the laboratory (microscopy, culture, identification, antibiotic sensitivity, serology, molecular tests).
3. **Post-analytical (post-test)** – reporting results to the clinician.

General Precautions for Collection

- Specimen should be **appropriate and representative** of the infection – e.g. **sputum, not saliva**, for pulmonary TB – with adequate **quantity** and quality.
- Collect **before starting antimicrobial therapy**.
- Collect **aseptically** to avoid contamination by skin or environmental flora.
- Use a **sterile** swab or container of adequate size; **anaerobic** container or transport medium for anaerobic culture.
- Label properly and send with a request form; the procedure should **not harm the patient**.

Transport and Storage

- Transport to the laboratory **without delay**; if delay is unavoidable, **store at the appropriate temperature** for a specified time.
- **Viruses** – send in **viral transport medium** (prevents drying, maintains viability, prevents overgrowth) held at **4°C**.
- Delicate organisms (gonococcus) – immediate transport; **modified Stuart's medium**.
- **Feces** – fresh sample; if delayed, **Cary–Blair** or **Venkatraman–Ramakrishnan (VR)** medium; for *V. cholerae*, **alkaline peptone water**.

Common Specimens

Skin (scrapings, swabs, biopsy), pus (aspirate), nail and hair clippings, conjunctival and ear swabs, throat/nasopharyngeal swabs, sputum, bronchoalveolar lavage, stool, vomitus, genital swabs and discharge, urine, suprapubic aspirate, **CSF, blood**, bone marrow, and serum/plasma for antigen or antibody tests.

Collection and transport of important specimens

Specimen	Collection	Transport / storage
Blood (culture)	Aseptic venipuncture before antibiotics ; adults 7–10 ml, children 1–5 ml; 3 samples in 24 h in endocarditis; inoculated into blood culture broth (BHI, trypticase soy).	Incubate at 37°C.
Urine	Clean-catch midstream urine (MSU) ; catheter specimen (CSU) with sterile syringe from catheter – never from the uro-bag ; suprapubic aspirate in infants; early morning urine × 3 days for renal TB.	Room temperature within ½ hour or 4°C up to 4 hours ; for longer delays, boric acid container (up to 24 h).

Specimen	Collection	Transport / storage
CSF	Lumbar puncture under asepsis into 3 sterile containers (cell count, biochemistry, culture).	For bacteria: room temperature ; never refrigerate – if delayed keep at 37°C . For viruses: on ice.
Throat swab	Two swabs from tonsillar pillars (smear + culture); part of membrane in diphtheria.	Send promptly.
Sputum	Deep cough sample (induced by hypertonic saline if needed); else endotracheal aspirate or BAL.	Send promptly.
Stool	Sterile wide-mouthed screw-capped container; rectal swab if needed; food and vomitus in food poisoning.	Cary–Blair / VR medium; APW for cholera.
Genital discharge	Sterile container or swab from base/edge of ulcer; capillary tube for vesicle fluid.	Immediate; modified Stuart's medium.
Pus / wound	Aspiration or incision-drainage preferred over swab.	Send promptly (anaerobic container if needed).

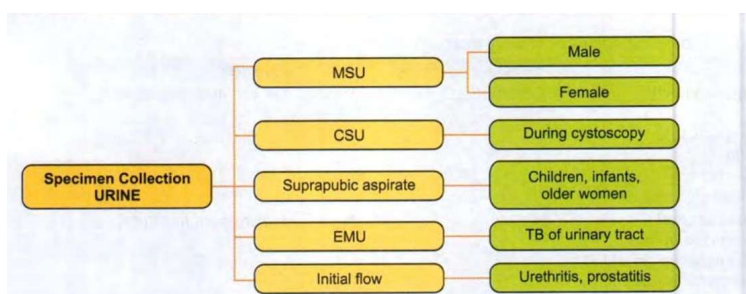


Fig. 76.1 Types of urine samples to be collected (MSU – midstream urine; CSU – catheter specimen; EMU – early morning urine) (from the book)

Processing and Reporting (in brief)

Processing begins with **microscopy** (Gram, ZN, KOH mount, fluorochrome stains), followed by **culture** on suitable media, **identification** (biochemical tests, serotyping) and **antibiotic sensitivity testing** (Kirby–Bauer). Rapid methods include antigen detection (latex agglutination, ELISA), molecular tests (PCR) and antibody detection (IgM/IgG ELISA, Western blot). Results must be reported accurately and clearly. **Quality control** is maintained in all procedures, and laboratories are accredited by **NABL** in India to ISO standards.

★ EXAM FOCUS

- Short notes: **Precautions for sample collection, Common samples received in the microbiology laboratory, Collection of urine / CSF / blood for culture.**
- Remember: CSF is **never refrigerated** for bacterial culture; urine – 4°C up to 4 h; viral samples – VTM at 4°C; stool – Cary–Blair.